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The Impact of Post-Training Techniques on Feature Representations

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Abstract

Upon completion of the initial training process, all LLMs used by us daily go through a number of methods designed to alter their behaviour and personality traits. These processes, which fall under the post-training umbrella, include Reinforcement Learning from Human Feedback (RLHF), Direct Preference Optimization (DPO), and more. However, the majority of mechanistic interpretability research has thus far been conducted in relation to the original base models, while the representational effects of post-training on instruction-tuned models have remained largely unexamined.

This thesis will ask if instruction tuning adds new features to a language model’s internal representation, or simply rearranges those already present within the original model. This question is addressed via a comparison between feature dictionaries produced by matched Sparse Autoencoders from the Gemma Scope 2 release [19], which contains feature dictionaries directly comparable between the pre-trained and instruction-tuned versions of the Gemma 3 model family. To address this question, a four-stage pipeline was created to collect residual stream activations from both checkpointed versions of Gemma 3 over a total of 5,000 FineWeb sequences. Activations were classified based on how they relate to each other according to both activation correlation and decoder cosine similarity. Those classifications were then related to descriptive labels derived from a frequency-weighted sample via dual-LLM consensus annotation. Finally, the resulting classifications were compared against differential activation test results generated from two behavioral comparisons, utilizing three probe datasets: a safety comparison on JailbreakBench, and an instruction-following comparison between Stanford Alpaca prompts and TruthfulQA as a factual reference point.

The proportion of cleanly preserved features relative to the entire set of active features turns out to be very low. Features related to instruction-shaping input and harmful content are primarily already represented in the base model; instruction tuning enhances and re-assigns these instead of adding new ones. As such, this represents the first feature-level evidence supporting the “wrapper hypothesis” of Jain et al. (2024), previously examined in synthetic contexts only. This suggests that instruction tuning rearranges a pre-existing representational substrate, rather than generating new representations.

1 Introduction

1.1 Problem and Motivation

All the modern large language models (LLMs) being utilized by millions of users today have had to go through extensive post-training pipeline modifications that involve layered supervised instruction tuning, preference learning from user feedback, Constitutional AI, and direct preference optimization on top of a base model that was initially trained to predict the next token based on Web-Scale Text [2, 6, 24, 30, 39]. It is primarily the post-training component, rather than the base model, that gives the deployed system its characteristic behaviours such as following the given instructions, responding using a certain tone/register, rejecting dangerous/inappropriate requests, and maintaining the role of an assistant. However, the majority of mechanistic interpretability research has focused on understanding base models. From Olah et al. [23]’s Zoom In article concerning circuits, to the indirect object identification circuit of Wang et al. [34], to the factual association edits of Meng et al. [20], and the feed forward key-value memory of Geva et al. [11]. Consequently, there is a gap between the layer of the language model that interpretability understands best and the one that users experience.

There are many reasons why this gap is important beyond just providing a complete picture. For example, without knowing how instruction tuning affects a model at a representational level we will never be able to determine whether a behavior exhibited by a deployed system represents a true property of the model or if it is simply an artifact created during post-training that would be easily eliminated through additional fine-tuning, could be ”jail broken” via adversarial prompt engineering, or even circumvented by deceptive alignment [12, 40]. Currently our ability to address these types of questions relies heavily upon behavioral analysis (probe the deployed system, observe whether it rejects dangerous/prompts, etc.) and therefore, an inside-the-box representation analysis of what internal representations post-training introduces/uses/reconfigures is missing.

A widely cited position in regards to the mechanism of Instruction Tuning is the Superficial Alignment Hypothesis presented by Zhou et al. [38]: surprisingly little amounts of high quality

instructional data are required to generate strong instruction following behaviour, suggesting that Instruction Tuning may not create new representations but instead teaches a model to access pre-existing ones. An opposing position that is less frequently referenced but is implied in much of the alignment literature suggests that Instruction Tuning creates new computational structures (features/circuits/determination processes) that did not previously exist in the base model. Both positions imply that Instruction Tuning creates different representations within the model and that distinguishing between them requires determining whether a specific feature present in the Instruction-Tuned Model’s Layer 13 Residual Stream has a corresponding feature in the Base Model’s Layer 13 Residual Stream.

Until relatively recently, addressing this difference at a representation level was practically unfeasible. The primary mechanisms available for performing mechanistic interpretability (classifier probing, activation patching, attention pattern inspection, residual stream decomposition) were either designed to measure overall behavioral properties of a model or focus on analyzing single circuits at a time; none of these mechanisms provided an approach for directly comparing the entire representation state of two models.

Sparse Autoencoders resolved this issue. Initially explored in terms of toy-model superposition (Elhage et al. [9]) and early results on monosemantics ([7]), sparse autoencoders now enable the creation of feature dictionaries describing residual-stream activations of a model. Each entry in the dictionary describes a vector pointing along a dimension of activation space tending to activate along semantically interpretable patterns. As such, two checkpoints representing the same model prior to and subsequent to post-training can be equipped with sparse autoencoders trained on the same residual-stream locations and have their feature dictionaries compared directly. A feature represented in both dictionaries that point in the same direction and activate on the same input has been preserved throughout post-training. A feature represented in only one dictionary or whose direction or functional behavior has been altered was modified, added or deleted. This type of comparison operationalizes the empirical dispute over whether Instruction Tuning creates new capabilities versus merely aligning existing capabilities towards instructional tasks. This is precisely what this thesis aims to investigate.

Why Study Instruction Tuning Specifically? Practically, instruction tuning is the first and

least complex stage in post-training; therefore its effect can be separated from those resulting from subsequent reinforcement learning-based stages, provided that matched Sparse Autoencoders exist for both the base model and the supervision-fine-tuned checkpoint. The Gemma Scope 2 Release ([19]) provides matching Sparse Autoencoders for both the Gemma 3 1B base and Instruction-Tuned versions; this constraint ultimately determined the selection of model for this thesis (See Subsection 2.1). Scientifically, Instruction Tuning is also the stage whose mechanism is most directly investigated by the superficial-alignment literature and very recent mechanistic fine-tuning work ([8, 13, 28]); thus, a feature-level investigation enables a connection to an ongoing debate rather than initiating one entirely.

1.2 Prior Art and Research Question

The focus of this section is a review of the prior literature which frames the research question. It is organized into thematic sections across four sub-subsections. Subsection 1.2.1 provides an overview of representational foundations: linear representation hypotheses, superpositions, and dictionary learning, as alternative solutions to the issue that individual neurons are not the right unit for analysis. Subsection 1.2.2 describes how sparse autoencoders have progressed from toy models to the Gemma Scope releases that enable this very research. Subsection 1.2.3 explores today’s knowledge of to the impact of post-training on internal representations, including the superficial alignment theory, mechanistic fine-tuning work, and representation engineering. Finally, Section 1.2.4, identifies the gap addressed in this dissertation and articulates the research question.

1.2.1 Representational Foundations: Superposition, Linear Features, and the Case for Dictionary Learning

It has been acknowledged that the unit-of-analysis issue in mechanistic interpretability has existed prior to the actual formation of the mechanistic interpretability area; i.e., the issue had already arisen in research on distributed word representations [21] which demonstrated that directions within an embedding space exhibit semantically meaningful structure which is uninterpretable based upon any one of the individual coordinates. Individual neurons are not the

proper unit of analysis since each neuron contributes to many unrelated computations. Radford, Jozefowicz, and Sutskever [29] illustrated this concept using character-level language models. Although Radford et al. found that there were exceptions, such as a "sentiment neuron", they noted that sentiment neurons would not typically be the norm. Olah, Mordvintsev, and Schubert [22] and Olah et al. [23] developed the overarching viewpoint that the appropriate unit of analysis should be the feature, a direction within activation space representing semantically legible patterns, which may or may not correlate with a particular neuron.

There exists a formal explanation that neurons do not constitute the proper unit of analysis due to *superposition* defined by Elhage et al. [9]: a network containing n number of neurons can only provide n orthogonal directions exactly; however, a network can provide many more than n near-orthogonal directions approximately if those near-orthogonal directions are sparse (inactive on most inputs). A sparse set of features can be compressed into a lower-dimensional space with tolerable interference. As a result, the representation provides each feature with a direction that is a linear combination of several neurons. Therefore, single-neuron interpretation is not only aesthetically limited but also methodologically incorrect when examining overcomplete learned representations: the neurons will be polysemantic by design. Scherlis et al. [32] further extends this analysis by developing a more rigorous account of polysemanticity as a function of capacity; Park, Choe, and Veitch [25] also develops the linear representation hypothesis, that meaningful features correspond to directions rather than nonlinear manifolds as structural claims about transformer internals, which can be tested.

Superposition motivates dictionary learning as the appropriate remedial action. If true representational units are features residing as sparse linear combinations of neurons, then the correct method for recovering them would be to learn a dictionary of feature directions and an encoder which assigns each activation a sparse code over said dictionary. The above idea is classical within signal processing. It was first demonstrated with language models using dictionary learning applied to neural networks in Cunningham et al. [7] who showed that dictionaries trained on sparsely activated transformer activations provided substantially more interpretable features than were obtainable through single-neuron interpretation. Simultaneous work in Anthropic's Towards Monosemanticity line [4] further developed the approach to larger transform-

ers and established much of the vocabulary (“features,” “monosemantic directions,” “activating examples”) now standard in the SAE literature. This thesis relies upon the premise that features recovered through dictionary learning on a transformer’s residual stream are meaningful representational units and that two transformers whose residual streams have been decoded in terms of dictionary learning in the same manner can be compared at the level of their features

1.2.2 Sparse Autoencoders as an Interpretability Tool

Early SAE research applied a form of the L_1 sparsity penalty to the output of the encoder, but it is well-documented that this penalty has well-known failure modes. These include the fact that when applying an L_1 penalty to active features, those features will be shrunk towards zero, thereby introducing a strong bias into both the reconstruction process as well as downstream analyses. In addition, because the sparsity regimes in early SAE formulations were fixed, tuning such regimes across different layers and/or different scales can be extremely challenging. Therefore two improvements have been developed. Rajamanoharan et al. [31] showed that by creating gated SAE’s one can disentangle the decision of whether or not to activate a particular feature from how much to activate that feature, thus greatly decreasing the degree to which the L_1 penalty shrinks the value of active features. Later, Gao et al. [10] found empirical scaling laws relating SAE performance as a function of three parameters: dictionary width, training compute and sparsity target size, that guided many of the subsequent SAE releases. Additionally, Karvonen et al. [15] created SAEBench, an evaluation benchmark that measures the quality of SAEs using more than just reconstruction loss (as measured via feature interpretability, probe-based downstream utility, etc., as well as robustness to perturbations).

The most relevant artefact for this thesis is the Gemma Scope 2 technical report and the associated matched-SAE release for Gemma 3 [19]. The original Gemma Scope suite included trained Sparse Autoencoders for every layer and several widths on Gemma 2 [17]; the documentation of Gemma Scope 2 described the paired base and instruction-tuned SAEs for Gemma 3, including the dictionaries that are being utilized. Due to the scale and reproducibility of the Gemma Scope releases it became possible for researchers to assume there existed high-quality trained SAEs for use with a production class model, whereas before these releases,

researchers would need to train their own. Peng et al. [27] states that SAEs are most beneficial for discovering new concepts rather than utilizing pre-existing knowledge of those concepts, which provides additional insight into the nature of the Targeted Analysis Stage of this thesis, where behavioral probe datasets allow the feature space to create its own concepts and do not require researcher-defined concepts.

1.2.3 Post-Training and Its Effects on Internal Representations

Post-training consists of all stages and processes that a language model undergoes after pre-training on next-token prediction. The typical pipeline begins with Supervised Instruction Tuning [35, 38]; followed by Reward Modeling [6, 39]; and then concluding with Reinforcement Learning from Human Feedback [24]. Constitutional AI [2], and other similar approaches substitute or add model-generated feedback based on explicit principles in place of human feedback, and Direct Preference Optimization (Rafailov et al. [30]) transforms the RL phase into a direct optimization problem, which circumvents the need for a reward model. The more general Helpful-and-Harmless perspective in Bai et al. [3] defines the objective as developing a model that is both useful and safe.

Previous research in the interpretability literature of how these actions affect the model’s internal representations comes from three different perspectives. The research question of this dissertation exists at their intersection.

The Superficial Alignment Argument. The most robust expression of this view is LIMA [38], where the principal result is that 1000 hand-crafted instruction-response pairs can be used to generate high-quality instruction-following behavior in a large base model, comparable to models fine-tuned over many orders-of-magnitude greater amounts of data. Their own interpretation is clear: a model’s knowledge and abilities are acquired almost completely through pre-training, while alignment simply tells the model which subset of formats to draw upon when interacting with users. This is an argument supported by behavioral evidence, and does not imply any specific mechanism of representation, however it makes a clear prediction regarding what an interpretability study would find: that the substrate for instruction-following behavior is present in the base model prior to fine-tuning; and fine-tuning only changes the

features that are read out and in what register, rather than creating new ones. LIMA does not test this prediction at the representational level; but frames the empirical question answered by this dissertation.

Mechanistic Fine-Tuning Line. In parallel, mechanistic research uses circuit-level and probe-level tools to look into what structures inside the model change when it is fine-tuned. The foundational piece of literature for this dissertation is Jain et al. [13], which studies the fine-tuning process in controlled synthetic contexts through the use of compiled Tracr transformer models and probabilistic context-free grammars. In particular, it demonstrates that the application of fine-tuning rarely results in changes to the underlying capabilities of a model. Rather, the process typically identifies a "minimal transformation" or "wrapper," on top of these capabilities. The wrapper hypothesis (the idea that fine-tuning instills a minimal re-routed version of the existing capabilities rather than creating entirely new capabilities) is the claim tested in this dissertation in a novel context. While Jain et al.'s support their findings through the use of pruning, probing, and reverse fine-tuning in synthetic environments with known ground truth, they do so without conducting feature-level analysis based upon SAE or utilizing a production-quality model. Therefore, extending their findings to the feature-level analysis conducted with a production-quality instruction-tuned LLM is the extension made in this study.

Immediately adjacent to the study conducted by Jain et al.'s, Prakash et al. [28] utilizes similar circuit-level tools to investigate how well fine-tuning for entity tracking actually modifies the behavior of the base model at the circuit level. Specifically, it demonstrates that fine-tuning does not create a new circuit within the model; however, it does enhance both the selectivity and magnitude of a single circuit that exists in the base model. Du et al. [8] builds upon this understanding of how fine-tuning affects circuits and seeks to determine which elements of a model's behavior are most significantly impacted by post-training at the representational level. Finally, Xu et al. [37] investigates the mechanisms underlying pre-training itself at a finer granularity and provides the additional insight that feature-level analysis can be utilized to monitor representational changes across training, providing a temporal analog to the checkpoint comparisons provided in this dissertation.

Representation engineering and the causal aspect. A third area of literature considers

representations both as subjects of interpretation and as objects of manipulation. Zou et al. [41] proposed a top-down strategy to extract concept directions from model activation by way of contrasting pairs and utilize those to observe and control behavior. They demonstrated that through simple linear transformations, one could interpret properties such as honesty, refusal and emotional valence from the residual stream, and manipulate them into it. Therefore, this represents the causal analog to correlational, feature-level analyses of SAE. While the literature surrounding representation engineering serves as background for this dissertation, its empirical contribution will remain limited to description and correlation, with causality being addressed in behavioral terms in subsequent work (discussed in Subsection 5.5).

The motivation behind the jailbreak and adversarial attack literature ([12, 40]) comes next. If safety behaviors are just a wrapper over the capabilities of a base model, then adversarial prompts that bypass the safety mechanisms should surface capabilities the instruction-tuned model would otherwise not have. A feature-level description of which representations are shared between the two checkpoints is directly relevant to understanding where such surfacing can and cannot reach.

Similarity-of-representations work. One further adjacent piece: Kornblith et al. [16] developed Centered Kernel Alignment (CKA) and related similarity metrics for comparing neural network representations. CKA computes a global metric that quantifies the similarity between representations produced by two neural networks at a particular layer. It has been applied widely to compare the representations produced by fine-tuned models relative to those of their respective base models. Aggregate similarity is complementary to the feature-level comparison of this research. CKA measures how different two representations are globally, while a feature-level comparison identifies specifically which dimensions are different and to what extent. This dissertation did not calculate CKA between the PT and IT residual streams; however, if desired, the feature level results from this study could potentially be checked against results obtained from calculating such a global similarity in future studies that replicate the findings presented within this dissertation.

1.2.4 The Gap and the Research Question

Literature surveyed in Subsubsections 1.2.1 through 1.2.3, indicates that the three major points that have emerged from this body of literature are: (i) sparse autoencoders have developed to a point where they provide a feature-level lexicon for the residual portion of a production-grade model; (ii) there exists matched SAEs for the Gemma 3 1B base and instruction-tuned checkpoints by way of Gemma Scope 2 [19]; and (iii), there is an empirical hypothesis which suggests that instruction tuning largely re-exposes previously installed base model capabilities instead of adding entirely new ones, notably expressed in LIMA and implied throughout much of the alignment literature. Further, this hypothesis has been empirically tested with other non-SAE-based tools in artificial environments and with circuit-level based tools in task-specific environments. The results have generally supported this hypothesis in each case. However, this hypothesis has not yet been examined using feature-level Sparse Autoencoders on actual instruction-tuned LLMs.

This thesis bridges the gap established by these findings: no previous research has used matched Sparse Autoencoders to examine the feature-level representations of a real pre-trained and instruction-tuned language model at the exact same layer, nor has it investigated whether the behaviourally relevant features of the instruction-tuned version are maintained, modified or completely novel compared to its base model. The ability to do so was made possible by the Gemma Scope 2 release [19] regarding the comparison of the pre-trained and instruction-tuned variants of Gemma 3 1B, and therefore motivated the pipeline outlined in Section 2.

The research question: at the feature level, does instruction tuning add new features to a language models' internal representations, or are they mainly reorganisation of previously present ones? More specifically, if matched Sparse Autoencoders are applied to decode the residual streams of a pre-trained and an instruction-tuned variant of Gemma 3 1B at layer 13, are the features which are most important to understanding the performance of a language model in terms of its ability to follow instructions and safety related prompts best described as being preserved, modified or unique to one of the two checkpoints?

This is a narrower question than the entire superficial alignment vs. new capability debate. It limits itself to a single layer of a single model pair, a single post-training phase and an

explicitly descriptive/correlational approach.

This study will be operationalized by two sub-questions:

- RQ1: The percentage of characteristics in Layer 13 of Gemma 3 1B that were kept, changed, PT-exclusive or IT-exclusive when comparing the base checkpoint to the IT checkpoint; and how robust is that percentage to changes in matching thresholds?
- RQ2: When the IT model is probed with safety-relevant and instruction-following prompts, do the features with the greatest differential activity correspond to IT-exclusive entries according to a strict creation interpretation, or to those features matched to an existing PT entry in the pipeline?

1.3 Main Contribution

This thesis contributes a four-stage empirical pipeline for testing whether instruction tuning introduces new features into a language model’s internal representations or predominantly reorganises pre-existing ones, together with the first application of this pipeline to a real instruction-tuned language model using matched Sparse Autoencoders. The pipeline is applied to the pre-trained and instruction-tuned variants of Gemma 3 1B at the residual stream of layer 13, using the `gemma-scope-2-1b-pt-res` and `gemma-scope-2-1b-it-res` SAEs from Gemma Scope 2 [19].

The concrete deliverables are three.

1. **The pipeline itself.** Four loosely coupled stages—activation collection, feature matching across checkpoints, dual-LLM consensus labelling, and targeted differential activation analysis—described in full in Section 2. Each stage is reusable independently with any pair of matched SAEs or any set of SAE features.
2. **Empirical findings on Gemma 3 1B.** The pipeline produces evidence that the features most behaviourally relevant to instruction-following and safety-related prompts are predominantly reorganised pre-existing features rather than newly introduced ones, a descriptive result consistent with the wrapper hypothesis of Jain et al. [13]. Full results

appear in Chapter 3; the reader should expect a nuanced picture in which cleanly preserved features are a small minority, the modified class dominates, and instruction-style features concentrate in the modified rather than the IT-exclusive class.

3. **A reusable methodological template.** This is the first published application of matched SAE comparison between a base and an instruction-tuned checkpoint of a production-class language model, defining a protocol reusable as matched SAEs become available for other model pairs.

What the thesis does not claim matters too. The evidence is correlational, not causal: Chapter 4 names full causal validation as the primary limitation and the most important direction for future work, and Chapter 5’s verdict on the wrapper hypothesis is therefore “consistent with” rather than “supports.” What the thesis does claim is the weaker position: at the residual stream of layer 13 in Gemma 3 1B, the features carrying the behavioural signatures of instruction tuning are predominantly reorganised from a pre-existing base-model feature population rather than introduced from scratch.

2 Methodology

This chapter defines the four-stage analytical pipeline used to test whether instruction tuning introduces new features into a language model’s internal representations or predominantly re-organises pre-existing ones, mapping directly onto the research question of §1.2.4. The focus is both methodological and operational: what is measured, how the four-way classification is constructed, and which concrete settings were used in this study. Stage one (Sections 2.1–2.2) fixes the model pair, interpretability tool, and activation corpus, so that the pre-trained and instruction-tuned variants are compared at the same architecture, layer, and SAE width using matched Sparse Autoencoders. Stage two (Sections 2.3–2.4) defines a feature-level correspondence between the two checkpoints based on decoder direction similarity and activation correlation, and uses a threshold grid to check that the resulting four-way partition (preserved, modified, PT-exclusive, IT-exclusive) is not an artefact of a single threshold choice. Stage three (§2.5) attaches descriptive vocabulary to sampled features through automated dual-LLM labelling with consensus construction, providing both the interpretive layer needed to discuss what features mean and an inter-annotator reliability check. Stage four (§2.6) grounds the classification in behavioural prompts by computing targeted differential activations for a safety contrast (JailbreakBench harmful versus benign) and an instruction-following contrast (Stanford Alpaca versus TruthfulQA) using Welch’s t -tests. All code was implemented in a single Jupyter notebook using SAELens and HuggingFace Transformers; the resulting artefacts are analysed in Chapters 3 and 4.

2.1 Stage 1: Model Selection and Sparse Autoencoders

The experimental subject is the Gemma 3 1B model family from Google: the pre-trained base checkpoint `google/gemma-3-1b-pt` and its instruction-tuned counterpart `google/gemma-3-1b-it`. Both share the same architecture (26 transformer layers, hidden size 1152) and were loaded at `bfloat16` precision using HuggingFace transformers. Gemma 3 was chosen over larger alternatives for two reasons. First, the 1B scale fits both variants, their matched SAEs, and the activation buffers for a 5,000-sequence corpus into a single commodity GPU, a hard constraint

for reproducibility and iteration speed. Second, this is the smallest scale at which publicly released SAEs exist for both a base and an instruction-tuned checkpoint of the same architecture, via Gemma Scope 2 [19]. Earlier project scoping targeted Gemma 2 2B, but that plan was abandoned because no matched SAE existed for the instruction-tuned checkpoint; without paired PT/IT SAEs at the same layer and width, the core comparison would not have been methodologically defensible.

Following the dictionary-learning framing of [4] and the formal setup of Cunningham et al. [7], a Sparse Autoencoder learns an overcomplete dictionary of feature directions $\{d_i\}_{i=1}^F$ in the residual stream space of a target transformer layer, together with a sparse encoder that assigns each activation $h \in \mathbb{R}^D$ a non-negative feature vector $f(h) \in \mathbb{R}^F$ such that $\hat{h} = \sum_i f_i(h) d_i$ approximates h in the mean-squared sense. The resulting feature directions are intended to correspond to semantically interpretable axes of variation, mitigating the superposition problem of Elhage et al. [9]. The specific SAEs used here come from Gemma Scope 2: `gemma-scope-2-1b-pt-res` for the base model and `gemma-scope-2-1b-it-res` for the instruction-tuned model, both at the `layer_13_width.16k.10_medium` configuration. Both have a dictionary width of 16,384 features, use the JumpReLU encoder of Rajamanoharan et al. [31], and were trained by Google DeepMind at comparable expected L_0 sparsity, making them directly comparable as feature dictionaries for the same architecture.

Layer 13 was chosen because it sits at the midpoint of the 26-layer stack, where prior work on Gemma Scope and residual-stream SAEs has reported the richest mixture of syntactic and semantic features. Middle layers are generally where abstract features relevant to downstream behaviour are most legible; earlier layers are dominated by token-level and syntactic features, later layers by output-oriented ones. This is a scoping decision, not a methodological claim, and its implications for generalisation are discussed in Chapter 4. One implementation detail worth recording: Gemma 3 1B required `bfloat16` rather than `float16` for inference, as the narrower dynamic range of `float16` produced overflow-driven NaN activations during early testing and would have silently corrupted the activation buffer.

2.2 Stage 1: Activation Collection

The activation corpus consists of 5,000 sequences drawn from the HuggingFace HuggingFaceFW/fineweb dataset (sample-10BT split), accessed in streaming mode with a fixed random seed for reproducibility. Each sequence was tokenised with the shared Gemma tokenizer and truncated to exactly 256 tokens; sequences shorter than 256 tokens were discarded rather than padded, so that every retained sequence contributes the same number of activation rows without requiring padding masks. The 5,000-sequence figure was downsampled from an initial 50,000-sequence plan after a pilot confirmed that the relevant SAE feature statistics (per-feature activation frequency, mean activation, per-sequence mean activation vectors used in §2.3) stabilised well below 50,000 sequences, and that additional data mainly increased runtime without changing the qualitative results. The final corpus contains approximately 1.275 million activation rows per model (5,000 sequences $\times$ 255 tokens after the first token of each sequence is dropped, as described below).

For each of the two models, activations were collected from the residual stream output of layer 13 using a PyTorch forward hook registered on the corresponding `model.model.layers[13]` module. Sequences were processed in batches of four, and within each batch the first token of each sequence was dropped before SAE encoding, because the first residual-stream position of a causal transformer carries no context from other tokens and its activation statistics are atypical. The resulting activation matrix was moved to CPU memory in `float32` and then re-encoded through the SAE in chunks of 64 rows at a time; this chunked encoding was necessary because naively encoding the entire activation matrix in a single GPU call caused out-of-memory errors on the 16,384-wide feature dictionary, and the one-sequence-at-a-time alternative was too slow to be practical. For each chunk, the SAE encoder, decoder, and reconstruction residual were computed, and three running statistics were updated per feature: a sum of activations, a sum of squared activations, and a count of non-zero activations. These running statistics were used after all batches to compute the per-feature mean activation $\bar{f}_i$ and activation frequency ρ_i . In parallel, a per-sequence mean activation vector was computed for each sequence, producing a matrix of shape 5000×16384 that was later used as the input to the

activation-correlation step of §2.3. Finally, two SAE quality metrics were tracked throughout encoding: the fraction of variance unexplained (FVU) of the reconstruction and the mean L_0 count, computed as the average number of non-zero features per activation row. These were reported at the end of each encoding pass as sanity checks that the pre-trained SAEs were operating within their expected regime on the FineWeb corpus rather than failing silently.

The output of this phase, for each model, is a summary of SAE feature behaviour over the corpus: activation frequency, mean activation, per-sequence mean activation vectors, and the SAE decoder matrices (loaded once and reused). These four objects are the only inputs required for the feature matching and classification stage described next.

2.3 Stage 2: Feature Matching and Four-Way Classification

The central methodological construct is the four-way classification of SAE features into preserved, modified, PT-exclusive, and IT-exclusive categories, built from two complementary similarity measures: decoder direction similarity, a purely geometric quantity computed from the SAE weights alone, and activation correlation, a functional quantity computed from how features co-activate on the shared FineWeb corpus. Neither alone is sufficient. Decoder similarity captures whether the two dictionaries point in the same directions in residual-stream space but ignores whether features with similar directions are actually used similarly by the two models. Activation correlation captures functional equivalence (whether a base-model feature and an IT-model feature light up on the same sequences) but depends on the similarity measure used for matching in the first place and is noise-dominated for rarely active features. Using both measures jointly, and requiring agreement between them for the strongest category (preserved), produces a classification robust to the weaknesses of either in isolation.

Decoder direction similarity is computed as follows. Let $D^{\text{PT}} \in \mathbb{R}^{F \times d}$ and $D^{\text{IT}} \in \mathbb{R}^{F \times d}$ denote the decoder matrices of the PT and IT SAEs respectively, where $F = 16,384$ is the dictionary width and d is the residual stream dimension. Both matrices are ℓ_2 -normalised row-wise so that every decoder direction is a unit vector, and the cosine similarity matrix is implicitly defined as $S = D^{\text{PT}}(D^{\text{IT}})^\top$. For each PT feature i , the best matching IT feature is found by taking $j^*(i) = \arg \max_j S_{ij}$ and its maximum cosine similarity is recorded as $s_i^{\max}$. The

same procedure is run in the reverse direction to find, for each IT feature, the best matching PT feature. This two-directional matching matters because the best PT-to-IT match is not in general the same as the best IT-to-PT match, and the direction of matching affects which feature of each pair “owns” the comparison. In practice the computation is done in chunks of 1,024 rows to keep the intermediate similarity tensor within GPU memory limits, but this is a performance rather than methodological detail.

Activation correlation is computed over the per-sequence mean activation matrices collected in §2.2. For each PT feature i and its best-matched IT feature $j^*(i)$, the Pearson correlation of their per-sequence activation vectors is computed over the 5,000 sequences. A small numerical guard is applied to avoid division by zero on features whose per-sequence activation vector has zero variance (which occurs for features that either never activate or activate at exactly constant magnitude). The resulting scalar $\rho_i \in [-1, 1]$ measures whether the PT feature and its geometric nearest neighbour in IT space are functionally correlated on the same inputs. The same quantity is computed in the reverse direction for IT features.

With these two measures in hand, the four-way classification at thresholds (τ_d, τ_a) is defined as follows. A PT feature i is called *preserved* if both $s_i^{\max} \geq \tau_d$ and $\rho_i \geq \tau_a$, meaning that its best geometric match in IT space is both close in direction *and* functionally correlated with it on the shared corpus. It is called *modified* if exactly one of the two conditions holds—that is, if the direction is preserved but the function has drifted, or the function is preserved but the direction has rotated. It is called *PT-exclusive* if neither condition holds, meaning that no strong geometric or functional counterpart exists in the IT dictionary. The fourth category, *IT-exclusive*, is defined symmetrically on IT features using the reverse-direction best matches and their activation correlations, and identifies IT features with no strong counterpart in the PT dictionary. Together, the preserved, modified, and PT-exclusive classes partition the active PT features, while the IT-exclusive class covers the complementary set of active IT features; a feature is treated as *active* if its activation frequency exceeds 0.1% of tokens, a threshold intended to remove dead features from the analysis without being restrictive enough to bias the category distribution. The active-feature threshold is reported explicitly alongside the classification results in Chapter 3 so that readers can distinguish distributional findings from artefacts

of feature pruning.

2.4 Stage 2: Threshold Selection and Sensitivity Analysis

The classification depends on two free parameters, τ_d and τ_a , and any single choice would leave the results vulnerable to the criticism that the reported proportions are threshold artefacts. To address this, the full classification is computed on a grid of 25 threshold pairs, with $\tau_d \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$ and $\tau_a \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$. For each pair, the number of preserved, modified, PT-exclusive, and IT-exclusive features is recorded, together with the preservation rate (preserved / active PT) and modification rate (modified / active PT). Results are visualised as three heatmaps (preservation rate, modification rate, PT-exclusive count) indexed by the threshold pair. The purpose of the grid is not to find an optimal pair (there is no ground truth against which “optimal” could be defined) but to confirm that the qualitative shape of the classification is stable across reasonable threshold choices.

The primary operating point for the interpretive phases (Subsections 2.5 and 2.6) is $\tau_d = 0.7$, $\tau_a = 0.3$, selected *after* the sensitivity grid was computed, on two grounds. First, it sits in the interior of the grid, so small perturbations in either threshold do not discontinuously change the class assignments. Second, it captures a moderate notion of both geometric and functional correspondence: $\tau_d = 0.7$ is strict enough to exclude near-random decoder alignments (for $d \sim 1000$, random unit vectors have expected cosine similarity close to zero, so 0.7 is well above chance) while being permissive enough to admit features whose directions have rotated mildly during fine-tuning; $\tau_a = 0.3$ tolerates activation noise from the small sample size while staying well above the near-zero correlations expected for unrelated feature pairs. Stricter settings such as $\tau_d = 0.8$, $\tau_a = 0.7$ collapse the preserved set and inflate the exclusive categories; more permissive settings wash out the distinction between modified and preserved. The chosen point is a compromise, and the full grid in §3.1 serves as the robustness check.

2.5 Stage 3: Automated Feature Labelling and Consensus Construction

A four-way classification without descriptions of what the features actually detect would be hard to interpret. To attach a descriptive vocabulary while keeping the effort manageable, this

stage samples 50 features from each category, for 200 labelled features in total, and submits their top-activating contexts to two independent LLMs, Claude Haiku 4.5 [1] and GPT-5.4-nano, for automated labelling. Using two models rather than one is deliberate: a single model’s labels are difficult to audit, while two independent models provide an inter-annotator reliability check and make labeller disagreement an observable that can itself be analysed.

Within each category, features are sampled with probability proportional to their activation frequency rather than uniformly. This choice weights the sample toward features that contribute meaningfully to the model’s computation over the corpus, and avoids allocating labelling budget to rarely-active features whose top-activating examples are noisy and hard to interpret. For each sampled feature, the top 20 activating token contexts are collected by running the appropriate model and SAE over the first 1,000 sequences of the 5,000-sequence FineWeb corpus used in Stage 1, extracting per-token feature activations, and keeping the highest-activating positions together with a window of 8 tokens on either side of the activating token. The activating token itself is marked with >>> and <<< delimiters in the extracted context so that the labelling model can unambiguously identify which token the feature fires on. This delimiter convention follows the display format used in the original Anthropic monosemanticity work and in subsequent SAE labelling pipelines.

Both models receive identical prompts and identical output schemas, so that any differences in output can be attributed to the models rather than prompt variation. The prompt instructs the model to act as a neural network interpretability expert, shows the 20 top-activating contexts for a feature, and asks for three outputs: a one-to-two sentence description of the detected pattern, a single-label category from a six-way taxonomy (`language_syntax`, `knowledge_semantics`, `instruction_style`, `safety_alignment`, `formatting`, `other`), and a confidence rating from 1 to 5. Output is constrained to this schema using the structured-output interfaces of the Anthropic and OpenAI SDKs (`client.messages.parse` with a Pydantic `FeatureLabel` schema for Haiku, `client.beta.chat.completions.parse` with the equivalent schema for GPT). Structured outputs were essential: an earlier free-text version produced inconsistent category strings (“syntactic”, “syntax”, “grammar”, “language”) requiring ad-hoc normalisation, while the schema-constrained version produces categories directly comparable across models. La-

labelling ran category-by-category with a rate-limit sleep between calls, and labels were serialised to disk as JSON after each category completed.

Consensus between the two labelling models is constructed as follows. For each sampled feature, if both models assign the same category, the feature is marked as an agreement case and that category becomes its consensus label. If the two models assign different categories, the feature is marked as a disagreement case, and the consensus label is taken from whichever model reported the higher confidence rating for that feature; in the event of equal confidence, Haiku is used as the primary model and GPT as the secondary. The consensus record for each feature additionally retains both descriptions and both confidence ratings, so that disagreement cases can be inspected qualitatively rather than only counted. The resulting consensus labels support two distinct uses in Chapter 3: first, the agreement rate and disagreement confusion matrix serve as an inter-annotator reliability diagnostic and give a measurable lower bound on the stability of the six-way taxonomy; second, within the agreement set, the distribution of categories across the four feature classes provides the descriptive evidence used to discuss what kinds of features are preserved, modified, or exclusive to each checkpoint. Both uses are discussed as results in §3.2.

2.6 Stage 4: Targeted Differential Analysis on Behavioural Probes

The labelling stage attaches descriptive vocabulary to features but does not by itself show that any of them matters for the behaviours instruction tuning is supposed to induce. To connect the classification to behaviour, a targeted differential analysis is run using two behavioural contrasts instantiated with three external probe datasets: JailbreakBench [5] for the safety contrast, Stanford Alpaca [33] for instruction-shaped prompts, and TruthfulQA [18] as the factual contrast set for the instruction-following comparison. In the concrete study reported here, the safety contrast uses 100 harmful and 100 benign JailbreakBench prompts, and the instruction contrast uses the first 200 Alpaca instructions against 200 TruthfulQA questions from the generation split. The contrast design matters. For safety, harmful-split JailbreakBench prompts are contrasted against benign-split prompts from the same dataset rather than against generic FineWeb text, so that the differential activation isolates safety-relevant structure rather than

incidental stylistic differences between task-oriented prompts and web text. For instructions, the Alpaca prompts are contrasted against TruthfulQA questions to produce a contrast between imperatively phrased task prompts and interrogatively phrased factual queries.

For each contrast and each of the two models, SAE feature activations are computed on every probe prompt using the same activation collection procedure as in §2.2, but with a shorter maximum sequence length of 128 tokens appropriate to the probe prompts and with per-prompt rather than per-sequence feature statistics. The per-prompt activation of a feature is defined as the mean of its token-level activations over all non-first tokens of the prompt, and is stored as a row in a prompts $\times$ features matrix per model. Two such matrices are produced per model—one for the safety-vs-benign contrast and one for the instruction-vs-truthful contrast—and all subsequent analysis is performed feature-by-feature on these matrices.

For each feature, a Welch’s two-sample t -test [36] is run comparing its per-prompt activations on the target group (e.g. harmful prompts) to those on the contrast group (e.g. benign prompts). Welch’s variant is used rather than the standard equal-variance t -test because the two groups cannot be assumed to have equal variance—a feature that is selectively active on harmful prompts is, almost by definition, more variable on the group where it activates than on the group where it does not. The implementation uses `scipy.stats.ttest_ind` with `equal_var=False`. Features are retained as differential if they satisfy two conditions simultaneously: a t -test p -value below 0.01 and a target-group mean activation above 0.01, the latter being a noise-floor filter that prevents the analysis from being dominated by features that are statistically elevated only because their baseline activation is essentially zero. For each retained feature the pipeline records its feature index, target-group and contrast-group mean activations, their difference and ratio, and the test statistic and p -value. The top 30 differential features per contrast per model are then ranked by difference and carried forward to the labelling pipeline of §2.5, which re-labels them on examples drawn from the probe prompts themselves rather than from FineWeb. This re-labelling step produces descriptions that are grounded in the behavioural contrast rather than in generic web text, and enables the results of §3.3 to discuss not only which features are differential but what they appear to detect.

A design note: the group mean activation reported for each differential feature is a con-

ditional mean, computed only over prompts in the target group, not an unconditional mean over the full prompt pool or FineWeb corpus. This means that comparisons across features or across contrasts based on the raw group-mean number reflect conditional rather than absolute activation strength. This is the appropriate choice for identifying features selective for a given behavioural domain (what matters is whether the feature lights up more on the target than on the contrast), but it must be stated explicitly when numbers are reported in Chapter 3 to avoid implying an unconditional baseline that was never measured. Chapter 4 returns to this point.

2.7 Implementation Notes

The pipeline was executed in Python inside a single Jupyter notebook on one GPU with approximately 79 GB of VRAM. Both Gemma 3 1B checkpoints were loaded in `bfloat16`; activation statistics and SAE encodings were accumulated in `float32` to match Gemma Scope training precision and avoid the overflow problems encountered with `float16` during early testing. The global random seed was fixed at 42. Intermediate outputs were written to disk after each stage, including the classification masks, threshold-grid summaries, raw annotator labels, consensus labels, and the targeted-analysis tables saved as `targeted_analysis_results.json`, so that later stages could be rerun without recomputing the earlier ones.

2.8 Summary

The four stages produce three artefacts feeding into the results and discussion of Chapters 3 and 4. The classification stage yields a four-way partition of the active SAE features in each checkpoint, together with a 25-point threshold sensitivity grid quantifying the partition’s stability. The labelling stage yields consensus descriptive labels for 200 sampled features, an inter-annotator agreement statistic, and a disagreement confusion matrix. The targeted stage yields two ranked lists of differential features (safety and instruction-following), with TruthfulQA serving as the factual contrast set for the latter, each annotated with Welch’s t -test statistics and classification status in the four-way partition so the two analyses can be cross-referenced at the feature level. Together these artefacts let Chapter 3 address the research question in descriptive, distributional, and behaviourally grounded terms, without committing to causal claims the

pipeline cannot support.

3 Results

This chapter reports the findings of the pipeline described in Chapter 2, organised around its three artefacts in increasing order of behavioural grounding. Section 3.1 reports the four-way classification and its threshold sensitivity. Section 3.2 reports the dual-LLM labelling results, including inter-annotator agreement and the distribution of semantic categories across feature classes. Section 3.3 reports the targeted differential analysis for two behavioural contrasts implemented with JailbreakBench, Stanford Alpaca, and TruthfulQA (the latter as the factual baseline for the instruction contrast), cross-referenced with §3.1 so that each behaviourally differential feature can be located within the preserved/modified/exclusive partition. Section 3.4 synthesises the three strands and states what they say about the wrapper hypothesis at the level of confidence the data warrants, which is descriptive and correlational rather than causal.

3.1 Feature Classification and Threshold Robustness

Activation collection on the 5,000-sequence FineWeb corpus produced reconstruction quality and sparsity statistics consistent with the Gemma Scope 2 SAEs [19] operating within their expected regime on both checkpoints. Both SAEs reconstruct the layer 13 residual stream with fraction of variance unexplained in the range reported by Lieberum et al. [17] for comparable `10_medium` configurations on Gemma 2, and both produce the anticipated long-tailed distribution of per-feature activation frequencies. The 0.1% activation-frequency cutoff from §2.3 retains a substantial majority of the 16,384 dictionary entries as active for each model. The active sets overlap considerably but not entirely: some features active on FineWeb for the PT model are effectively dormant for the IT model, and vice versa. This asymmetry is itself a first weak indication that the two checkpoints are not using their dictionaries identically.

At the primary operating point ($\tau_d = 0.7$, $\tau_a = 0.3$), the classification partitions the active feature set into four classes with telling proportions. Among active PT features, 241 are *preserved*, 4,017 are *modified*, and 3,884 are *PT-exclusive*; on the IT side, 1,899 active features are *IT-exclusive*. The *preserved* class (features whose best PT-to-IT decoder match both points in a closely aligned direction and correlates functionally on the shared corpus) is by far the

smallest of the three PT-side classes. The *modified* class (features satisfying one of the two criteria but not the other) is substantially larger. The *PT-exclusive* class (features with neither a geometric nor functional counterpart in the IT dictionary) is the largest. On the IT side, the *IT-exclusive* class covers a sizeable minority of active IT features. The threshold sensitivity grid that follows shows that this pattern is not specific to a single threshold choice. The qualitative reading: cleanly preserved features are the exception, and most active PT features either see their direction rotate, their function drift, or their counterpart disappear entirely when the model is instruction-tuned.

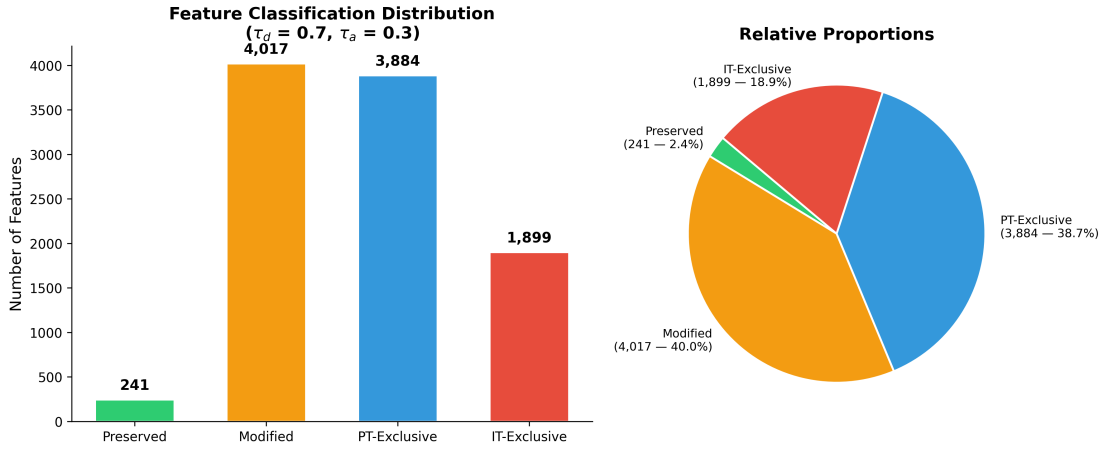


Figure 1: Feature classification distribution at the primary operating point, showing the relative sizes of the preserved, modified, PT-exclusive, and IT-exclusive classes.

The threshold sensitivity grid over the 25 pairs $(\tau_d, \tau_a) \in \{0.5, 0.6, 0.7, 0.8, 0.9\} \times \{0.1, 0.2, 0.3, 0.5, 0.7\}$ confirms that this shape is not an artefact of the chosen operating point. The preservation rate decreases monotonically as either threshold tightens, falling from a modest maximum at the most permissive corner to a very small fraction at the strictest corner, with no discontinuous jumps. The PT-exclusive count moves in the opposite direction, growing smoothly. The preservation rate never rises to a level that would sustain a preservation-dominated reading under any threshold setting: even at the most permissive corner, preserved features are a minority. The grid is reported as three heatmaps (preservation rate, modification rate, PT-exclusive count) in Figure 2; the row-and-column monotonicity visible in those heatmaps is the robustness evidence.

A secondary observation: the modified class is the largest across almost all threshold set-

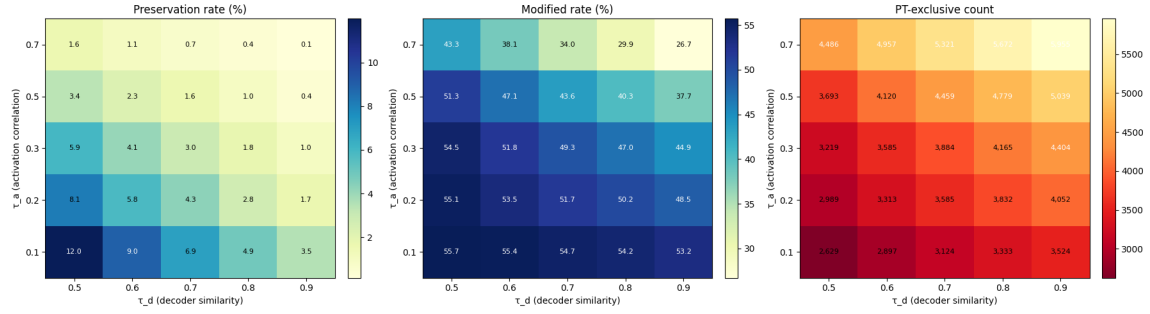


Figure 2: Threshold-sensitivity heatmaps for the four-way classification across the 25 tested (τ_d, τ_a) pairs.

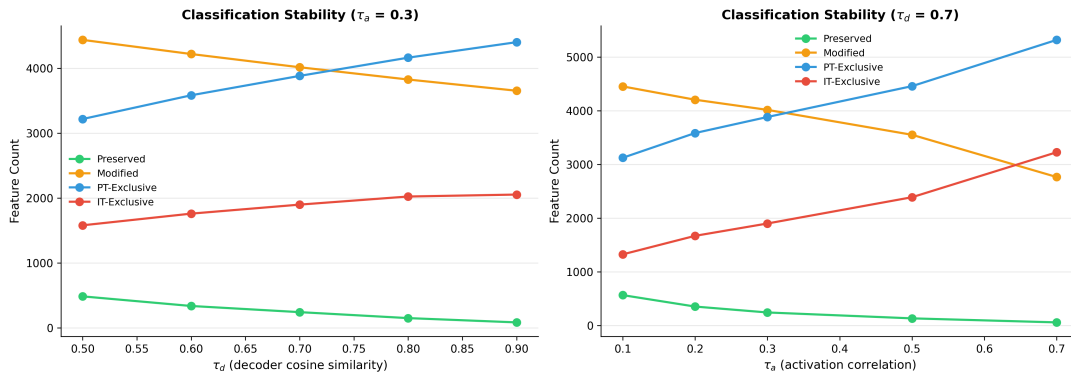


Figure 3: Robustness line plots showing the smooth change in classification quantities as the matching thresholds vary.

tings, with only the most extreme strict-corner combinations demoting it. The dominant relationship between PT and IT features at layer 13 is not replacement (“the IT model has new features PT lacked”) or preservation (“the IT model uses PT features unchanged”) but modification: features whose direction and function have drifted enough to fail one correspondence criterion while clearly retaining a counterpart on the other. The dominant mode of change is gradual reshaping of an existing feature basis. This matters in §3.4 when the classification is combined with the behavioural evidence.

In sum: at the primary operating point and across the full threshold sensitivity grid, preserved features are a small minority, modified features dominate, and the classification is robust to reasonable perturbations of both threshold parameters.

3.2 Automated Labelling and Inter-Annotator Agreement

The labelling procedure of §2.5 was run over 50 features from each of the four categories, for 200 labelled features in total and 400 label instances overall. Both Claude Haiku 4.5 and GPT-5.4-nano returned valid structured-schema outputs for every sampled feature. Self-reported confidence ratings are high on both sides: the mean confidence is comfortably above the midpoint of the 1–5 scale for both models, and category disagreement is not strongly predicted by low confidence. The labellers are often confidently disagreeing rather than hedging on ambiguous cases.

The two models assign the same category to 120 of 200 features (60.0% agreement), leaving 80 disagreement cases resolved by the confidence-weighted tiebreak of §2.5. The structure of disagreements is more informative than the headline number. Agreement is highest on preserved and modified classes (34/50 and 35/50, or 68% and 70%) and lowest on PT-exclusive (22/50, 44%), with IT-exclusive intermediate (29/50, 58%). The features easiest to label consistently are those with clear cross-checkpoint counterparts, whose activating contexts give both models evidence about what the feature detects. The hardest are checkpoint-exclusive features whose top-activating contexts are narrower and often capture idiosyncratic patterns specific to one model’s training distribution.

The disagreement confusion matrix shows that nearly all disagreement concentrates at

three category boundaries: `language_syntax` $\leftrightarrow$ `instruction_style`, `language_syntax` $\leftrightarrow$ `formatting`, and `knowledge_semantics` $\leftrightarrow$ `instruction_style`. The `language_syntax` row shows 25 features Haiku labelled as syntactic but GPT labelled as instruction-style, and 15 that Haiku labelled syntactic but GPT labelled as formatting. The reverse pattern (GPT choosing `language_syntax` where Haiku chose `instruction_style` or `formatting`) also occurs but at lower volume. The other three categories (`knowledge_semantics`, `safety_alignment`, `other`) show near-zero disagreement. The taxonomic uncertainty is not uniformly distributed but localised at the boundary between low-level linguistic form (syntax, punctuation, formatting tokens) and the features that respond to instruction-shaped prompts. The reason is straightforward: instruction-style features are often syntactically marked (imperative verb forms, infinitive constructions, sentence-initial directives), and the line between “this feature detects a syntactic pattern that happens to occur in instructions” and “this feature detects an instruction that happens to be syntactically marked” is genuinely hard. This is not a pipeline failure but a measurement of how fuzzy the categorical boundaries actually are, and is itself a finding.

Restricting to the 120-feature agreement set gives a cleaner view of the category distribution across classes. `language_syntax` is the largest category overall (70 of 120), followed by `knowledge_semantics` (31), `formatting` (16), and `instruction_style` (3). The near-absence of `safety_alignment` and `other` labels means that neither model, drawing from the frequency-weighted feature pool, found many clear safety-aligned or unclassifiable features. Two distributional observations matter. First, within the preserved class, `language_syntax` dominates almost entirely (28 of 34 agreed features), confirming that the features most stably preserved across instruction tuning are low-level linguistic and formatting features, not task- or content-level ones. Second, `instruction_style` is absent from the preserved class (0 of 34) and PT-exclusive class (0 of 22) in the agreement set, appearing only in the modified class (1) and IT-exclusive class (2). This small-count observation foreshadows a larger finding in §3.3: clear instruction-style features are a product of instruction tuning nearly absent from the base model’s most frequent active features, but they do not appear as large numbers of new features in the IT-exclusive class. Instead, they appear primarily inside the modified class, consistent with instruction tuning reshaping existing features into instruction-sensitive ones rather than

creating them from scratch.

In sum: inter-annotator agreement is moderate at 60%, concentrated on preserved and modified features. Disagreements are structurally localised at the syntax/instruction-style boundary, which is itself a finding about how fuzzy these categories are. Within the agreement set, preserved features are dominated by low-level linguistic categories, and instruction-style features appear almost exclusively in the modified and IT-exclusive classes. This descriptive observation needs the targeted analysis of §3.3 to be interpreted in behavioural terms.

3.3 Targeted Differential Analysis on Behavioural Probes

The targeted differential analysis combines the classification of §3.1 with the two behavioural contrasts of §2.6: safety (JailbreakBench harmful versus benign) and instruction-following (Stanford Alpaca [33] versus TruthfulQA, where TruthfulQA supplies a factual-question baseline rather than defining a separate truthfulness probe). Welch’s t -tests identify features whose per-prompt activation is significantly and substantially elevated on the target group relative to the contrast group. The full top-30 ranked lists per contrast per model are in `targeted_analysis_results.js`; this section discusses the top features and the structural pattern across contrasts.

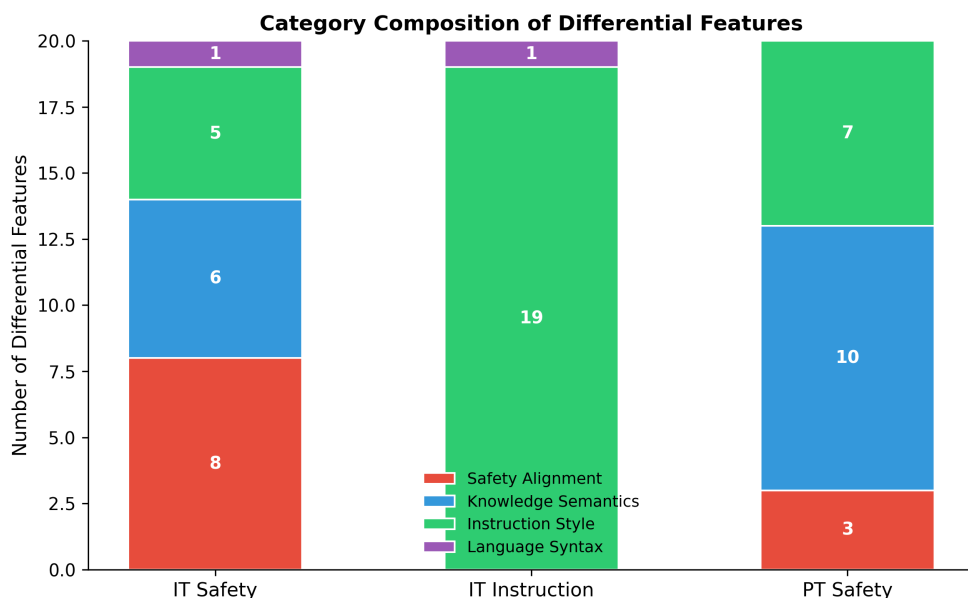


Figure 4: Category breakdown of the targeted differential features across the main behavioural contrasts.

Instruction-following contrast. The most striking result is in the instruction-following

contrast on the IT model. Of the 20 top differential features labelled for IT on Alpaca-versus-TruthfulQA, 19 receive the `instruction_style` category from the consensus procedure; the twentieth is `language_syntax`. The 19 instruction-style features are semantically coherent: feature 222 detects the opening of a quoted instruction block, feature 245 activates on imperative prompt-introduction verbs, feature 446 fires on task-initiating imperatives like “Give / Propose / Generate”, feature 2006 activates on output-transformation task markers, feature 374 completes common instructional scaffolds, feature 264 detects directives to adjust writing style, feature 3815 activates on the “the following” phrase pattern anchoring many instruction templates, feature 479 fires on enumeration-producing instructions, feature 793 detects short-output-length specifications, and so on. This is an essentially saturated result: the top differential features under an instruction contrast form a dense, semantically legible cluster, and the labelling procedure recovers it with near-perfect consistency. The statistical signal is overwhelming: top-ranked features have *t*-test *p*-values on the order of 10^{-33} to below floating-point precision, with activation ratios from roughly $10\times$ to over $140\times$ between Alpaca and TruthfulQA prompts.

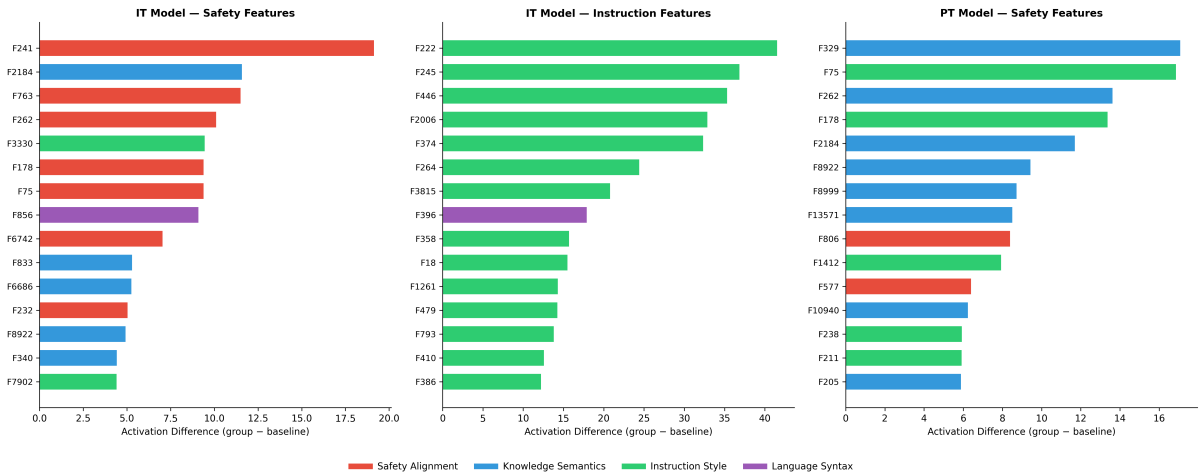


Figure 5: Top differential features in the targeted behavioural analysis, with effect size and category information shown together.

The question is whether this cluster is exclusive to the IT model or has counterparts in the PT model. Cross-referencing feature indices with the PT instruction-following differential list gives a direct answer. Two of the five top-ranked IT features (feature 2006 and feature 446) also appear in the top five of the PT list under the same contrast, with large effect sizes:

feature 2006 shows a PT activation difference of +114.3 and a ratio of 36.8 $\times$; feature 446 shows a PT difference of +37.5 and a ratio of 9.6 $\times$. Both are labelled `instruction_style` in the consensus and have clearly interpretable descriptions as instruction-shaped patterns. The interpretation is clear: the features that respond most strongly to instruction-shaped prompts in the IT model are, in substantial part, already present and selective in the base model. Instruction tuning increases their activation magnitude and sharpens their selectivity but does not create them. Locating these shared-index features in the §3.1 classification confirms that they belong to the modified class, not the IT-exclusive class: the pipeline independently identifies them as features whose direction or function has drifted, not features that appeared *de novo*.

Safety contrast. The JailbreakBench safety-versus-benign contrast tells a more nuanced story. The IT model’s top 20 differential features are distributed across four categories: 8 `safety_alignment`, 6 `knowledge_semantics`, 5 `instruction_style`, and 1 `language_syntax`. The PT model’s top 20 under the same contrast are differently weighted, with 10 `knowledge_semantics`, 7 `instruction_style`, and 3 `safety_alignment`. Two observations follow. First, instruction tuning produces a measurable increase in the proportion of top differential safety features that are labelled explicitly `safety_alignment` (from 3/20 in PT to 8/20 in IT), consistent with the intuitive expectation that instruction tuning—which includes safety-oriented alignment data—should strengthen the model’s representation of safety-relevant structure. Second, and more important for the wrapper reading, even the PT model has features in its top-20 differential list that are selective for harmful content: features labelled `knowledge_semantics` like feature 329 (wrongdoing completion/intensification), feature 262 (concrete harmful-action verbs), feature 2184 (malicious intent / operational action), feature 8999 (deceptive framing), and feature 13571 (coercion and manipulation) all show significant differential activation on harmful prompts in the base model.

Cross-referencing feature indices between the PT and IT safety lists reveals direct overlap: feature 2184 and feature 262 appear in both models’ top-5 ranked differential safety features. Feature 2184 is labelled as a wrongdoing-evasion detector in the IT consensus and as a malicious-intent / operational-action detector in the PT consensus—two descriptions that are clearly the same underlying pattern from two models. Feature 262 is similarly interpretable in

both cases. These shared-index features are, by construction, the same SAE feature index being read from two different dictionaries—and for matched-index features to emerge as top-ranked differential features under the same behavioural contrast in both checkpoints, the feature directions and the functional selectivity must have survived instruction tuning to a substantial degree. Locating these features in the §3.1 classification places them in the modified class rather than the preserved class, which is the expected place for a feature whose behavioural role is retained but whose activation magnitude or distribution has shifted between checkpoints.

The safety contrast is therefore weaker than the instruction contrast but consistent with the same reading: the features that light up on harmful prompts in the IT model are substantially drawn from the same set as those in the PT model. Instruction tuning reweights and refines the selectivity of existing features rather than introducing a new safety-dedicated cluster from scratch. The reweighting is real (the 3-to-8 shift in explicit `safety_alignment` labels is not noise) but it is a reweighting of which features dominate the top-20, not a categorical replacement. No feature in the IT top-20 safety list belongs to the IT-exclusive class; all have either a preserved or modified classification, meaning every one has at least some counterpart in the PT dictionary.

Cross-contrast observation. Two feature indices, 262 and 2184, appear in the top differential list for both the safety contrast and (in the wider top-30) the instruction contrast, across both models. These features fire selectively on both harmful content and instruction-shaped prompts, consistent with much of the “safety-relevant” structure at layer 13 being encoded not as a dedicated safety signal but as an intersection of instruction-following structure and semantic content about wrongdoing. This is a non-trivial observation about how safety information is represented in Gemma 3 1B at layer 13 and reinforces the reading that instruction tuning modulates the combination of pre-existing features rather than installing a dedicated safety circuit.

In sum: the instruction contrast produces an essentially saturated wrapper-consistent result. The top differential instruction-style features in the IT model are overwhelmingly already present and selective in the base model; instruction tuning amplifies rather than creates them. The safety contrast is weaker but consistent: instruction tuning reweights which features dom-

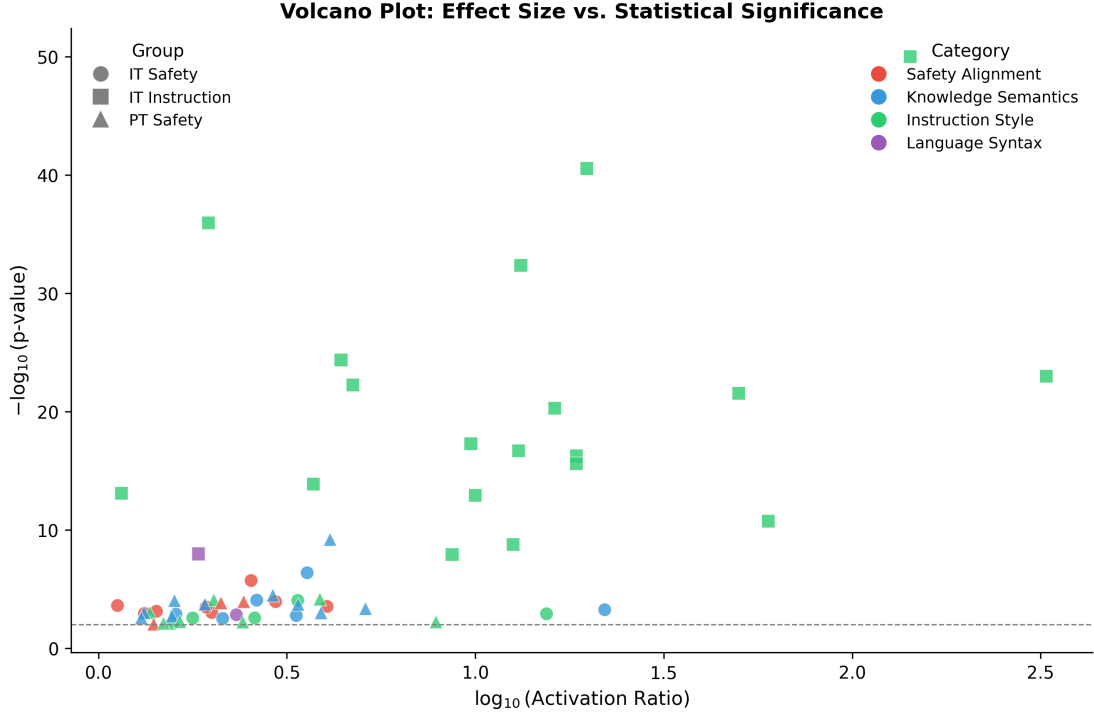


Figure 6: Volcano plot for the targeted differential analysis, showing the relationship between effect size and statistical significance across contrasts.

inate the top-20 toward more explicitly safety-aligned labels, but the features are substantially shared between checkpoints, and no IT-exclusive feature appears in the top-20 safety list. Both contrasts locate the behaviourally relevant features primarily in the modified class.

3.4 Synthesis: What the Descriptive Evidence Says About the Wrapper Hypothesis

The three strands of evidence converge on a consistent picture. The classification (§3.1) shows that cleanly preserved features are a small minority, modified features dominate across the full threshold grid, and a substantial fraction of active features in both checkpoints lacks a strong counterpart in the other, though this last fact does not translate into behavioural novelty of the IT-exclusive class. The labelling (§3.2) confirms that preserved features are overwhelmingly low-level linguistic rather than task-level, that instruction-style features are absent from the preserved class and concentrated in the modified class, and that the syntactic/instruction-style boundary is genuinely fuzzy even for capable annotators. The targeted analysis (§3.3) connects classification to behaviour: the features dominating the IT model’s response to instruction-

shaped prompts are largely the same indices dominating the PT model’s response, with instruction tuning amplifying selectivity by an order of magnitude or more; and the features dominating the IT model’s response to harmful prompts are substantially drawn from the same base-model population, with instruction tuning reweighting which features top the list rather than replacing them.

These findings are consistent with the wrapper hypothesis of Jain et al. [13] as originally formulated for procedurally defined tasks: the behaviours introduced by instruction tuning at layer 13 of Gemma 3 1B appear to be carried by reorganisation and reweighting of pre-existing features rather than creation of new ones. The strongest version (that no new features are created) is not supported: the IT-exclusive class is not empty. What is supported is the weaker claim that the features most behaviourally relevant to instruction-following and safety are not in the IT-exclusive class but in the modified class, whose members by construction have counterparts in the base model. The IT-exclusive class exists; it is just not where the behavioural action is.

Three caveats, developed in Chapter 4. First, the evidence is descriptive and correlational: it shows differential activation on behaviourally meaningful prompts but not that those features causally drive the behaviours. Second, the analysis covers a single middle layer of a single small model on a single corpus; none of these restrictions can be lifted from within the pipeline. Third, the labels depend on an automated six-way taxonomy whose boundaries are fuzzy (§3.2), and `instruction_style` and `safety_alignment` labels are not ground truth. With those caveats, the evidence is sufficient to conclude that a wrapper-style reading of instruction tuning is consistent with feature-level data from a real instruction-tuned language model, and to motivate the questions about causal validation and generalisation discussed in Subsection 5.5 and Section 5.

4 Limitations and Robustness

A descriptive pipeline is only as trustworthy as its boundary conditions are clearly stated. This section sets out what the study did not do, what it could not do, and what it did to check robustness. It moves from narrowest to broadest. Subsection 4.1 identifies the scope limitations fixed by design: single model pair, single layer, single corpus, 5,000-sequence sample size. Subsection 4.2 turns to measurement limitations internal to the pipeline: the conditional-mean baseline in the targeted analysis, the decoder unit-norm convention and its consequences for amplitude comparisons, and the reliance on automated annotators with fuzzy taxonomic boundaries. Subsection 4.3 reports a cross-SAE direction robustness check testing whether learned decoder directions behave differently from random directions of matched magnitude; this is the demoted remnant of a larger activation-steering experiment whose results did not support a clean behavioural interpretation, and this section explains both the surviving check and why the larger experiment is not reported as a primary result. Subsection 4.4 recaps the robustness checks Chapter 3 contains (threshold sensitivity, dual-annotator agreement, multiple behavioural contrasts, the cross-SAE check) and argues that together they define a defensible envelope of confidence around the thesis’s descriptive conclusions.

4.1 Scope Limitations

Single model pair. The pipeline operates on a single pair of checkpoints (PT and IT variants of Gemma 3 1B) and nothing else. Nothing in Chapter 3 tests whether the wrapper-consistent reading extends to larger models in the Gemma 3 family, to other model families (Llama, Mistral, Qwen), or to fine-tuning regimes other than instruction tuning, such as RLHF [6, 24], constitutional AI [2], or direct preference optimisation [30]. This restriction follows unavoidably from the matched-SAE requirement of §2.1: at the time of scoping, Gemma 3 1B was the only model with publicly released SAEs for both a base and instruction-tuned checkpoint at the same layer and width. Generalisation would require training SAEs for other models or waiting for matched community releases. The results should be read as a case study, not a general claim about instruction tuning.

Single layer. Activations were collected only from layer 13, the midpoint of the 26-layer architecture. The justification in §2.1 is defensible but not exhaustive. Layer 13 is a sampling point, not an optimum, and there is no a priori reason to expect the Chapter 3 results to be layer-invariant. Earlier layers should shift the feature distribution toward low-level syntactic and tokenisation features; later layers toward more task- and output-oriented ones. If the wrapper hypothesis holds layer-by-layer, the qualitative conclusion (modification dominates, behaviourally relevant features are shared) should also hold elsewhere, but quantitative proportions may differ. Multi-layer replication is discussed as future work in Subsection 5.5.

Single corpus and probe-dataset choice. Classification was run on a 5,000-sequence FineWeb corpus (sample-10BT split) [26], while labelling contexts were drawn from the first 1,000 sequences of that corpus; the targeted analysis used JailbreakBench, Stanford Alpaca [33], and TruthfulQA. Both choices introduce distributional biases. FineWeb is English-only web text and does not contain multilingual content, code, mathematical notation, or dialogue in significant fraction; features specialised for these domains will appear as low-frequency or dormant and be filtered by the 0.1% cutoff. Those three datasets implement two contrasts (harmful versus benign on JailbreakBench; instruction-shaped Alpaca prompts versus factual TruthfulQA questions), not three independent probe designs, and each contrast arm draws from a single source; the activation profiles of §3.3 may not transfer to other safety benchmarks (HarmBench), instruction datasets (Dolly, FLAN), or other factual-baseline choices than TruthfulQA. The choices should be read as representative, not exhaustive.

Corpus size. The 5,000-sequence corpus was chosen after a pilot showed that feature statistics (activation frequency, mean activation, per-sequence mean activation vectors) stabilised well below 50,000 sequences. This justification is sound for the classification stage, where what matters is the relative ordering and stability of per-feature statistics. It is weaker for labelling, where top-20 activating contexts are drawn from only the first 1,000 sequences of that 5,000-sequence FineWeb corpus, and a larger pool might sharpen labels for rare features. It is weakest for the targeted analysis, where the probe datasets are much smaller (100 harmful, 100 benign, 200 Alpaca, 200 TruthfulQA prompts) and the effective sample size for Welch’s t -tests is driven by per-group prompt counts. The 100–200 per-group counts are adequate for the large

effect sizes in §3.3 ($10\times$ to $140\times$ ratios, p -values below machine precision) but marginal for smaller effects. Targeting smaller-effect features would need a substantially larger probe set.

4.2 Measurement Limitations

Conditional-mean baseline in the targeted analysis. The group mean activation for each differential feature in §3.3 is a conditional mean computed only over the target group (e.g. harmful prompts), contrasted against a conditional mean over the contrast group (e.g. benign prompts). It is not an unconditional mean over a full background distribution. This is the right choice for identifying features selectively elevated on one group relative to another, but the consequence must be stated: the raw activation numbers in §3.3 are not absolute magnitudes. A feature with group mean 20 and contrast mean 2 is not necessarily “more active overall” than one with group mean 5 and contrast mean 0.05; the ratios ($10\times$ and $100\times$) describe selectivity between groups, not overall activation magnitude. An earlier draft of the pipeline was vulnerable to this confusion, and the distinction between conditional and unconditional baselines is a subtlety that careful readers of §3.3 will ask about.

Gemma Scope 2 decoder unit-norm convention. The Gemma Scope 2 SAEs [19] pre-normalise decoder rows to unit ℓ_2 norm. This is convenient for geometric comparisons (cosine similarity in §2.3 becomes a direct inner product) but has a subtle consequence for amplitude-based analyses: because all decoder rows have the same norm, a feature’s “strength” in the residual stream is entirely encoded in its scalar encoder activation, with no contribution from the decoder side. This does not affect the §2.3 classification or §2.5 labelling, which treat features as directions-plus-activations. It does affect any analysis comparing “natural” injection magnitudes across features, because a fixed-multiplier intervention ($h \leftarrow h + \alpha d_i$) will produce sub-threshold perturbations for features with typical activation much larger than α and over-large perturbations for features with typical activation much smaller. The cross-SAE check in §4.3 addresses this by using norm-calibrated multipliers based on empirical conditional mean activations.

Reliance on automated annotators. The descriptive vocabulary in §3.2 comes entirely from two automated LLM annotators, Claude Haiku 4.5 and GPT-5.4-nano. Manual annotation

of 200 features would have been prohibitively time-consuming and would not have provided the inter-annotator reliability check the dual-LLM setup gives for free. The cost is that the labels are not ground truth. The category assignments depend on a six-way taxonomy fixed before the labelling pass, and whose boundaries turn out to be genuinely fuzzy for both annotators in predictable ways: the `language_syntax` $\leftrightarrow$ `instruction_style` boundary alone accounts for a large fraction of disagreements. The 60% agreement rate is a defensible lower bound on taxonomic stability but not high enough to treat individual labels as reliable. Chapter 3 therefore uses categories distributionally (comparing proportions across classes) rather than resting claims on any single feature’s assignment. The per-feature free-text descriptions are used only qualitatively in §3.3.

Active-feature threshold. The 0.1% activation-frequency cutoff used to exclude dead features is a free parameter not subjected to its own sensitivity grid. The choice removes features that essentially never activate on FineWeb, which would contribute noise to correlation computation and waste labelling budget. A stricter cutoff would shrink the active set and bias toward the most frequent features; a looser one would admit rarely active features with noise-dominated statistics. Informal checks during development suggested that classification proportions are not strongly sensitive within the 0.05%–0.5% range, but no formal grid was computed. Extending the sensitivity grid of §2.4 to include this threshold as a third axis would be a straightforward extension.

4.3 Cross-SAE Direction Robustness Check

The Chapter 3 findings rely on two assumptions about SAE decoder directions worth testing separately. First, that these directions are not arbitrary but correspond to structured axes of variation in the residual stream. Second, that a decoder direction learned on one checkpoint retains structural meaning when applied to the other, which is what justifies the cross-model classification of §2.3. Both can be examined by a simple intervention experiment: inject a real SAE decoder direction into a model’s residual stream and compare the resulting distributional shift to the shift from a random unit vector of the same magnitude. If learned directions are structured in a way that matters for computation, real and random directions should produce

measurably different effects.

The experiment was implemented as a small activation-steering procedure at layer 13. For each test feature drawn from the differential lists of §3.3, the SAE decoder row was scaled by a norm-calibrated multiplier (derived from the feature’s empirical conditional mean activation on the relevant probe group, to avoid the unit-norm problem of §4.2) and added to the residual stream during the forward pass on probe prompts. KL divergence between steered and unsteered output distributions was computed across a small batch of prompts. The same procedure was repeated with a random unit vector in place of the real decoder direction, scaled to the same ℓ_2 norm, producing a matched-norm random control. A structurally meaningful decoder direction should produce a distributional shift systematically different from a random direction of the same magnitude.

The result, across test features from Case A (IT-exclusive directions injected into the base model) and Case B (PT-exclusive directions injected into the IT model), is consistent and statistically significant but in an unexpected direction. Real SAE decoder directions, injected at matched norm, produce *smaller* KL shifts than random directions of the same norm, with real-to-random ratios from approximately 0.5 down to below 0.1, and Welch’s t -test p -values reaching below 0.001 for the clearest cases. The model’s output is more perturbed by a random perturbation than by a structured one of the same size. This is statistically significant (the gap is robust, repeatable, and large) but it does not admit the behavioural reading a steering experiment would normally target. Interpretable evidence for or against the wrapper hypothesis would require the real direction to produce a larger, directional, behaviourally interpretable shift than random, and none of those conditions is met.

The most natural interpretation is structural rather than behavioural: the model “absorbs” interventions along its own learned feature directions more gracefully than arbitrary perturbations, consistent with these directions corresponding to axes of variation the model already expects and has machinery to handle. A random direction of the same magnitude is more disruptive because it pushes the residual stream off the manifold of activations the model was trained to process. This reading is consistent with the observation that SAE features correspond to structured directions in residual-stream space [7, 25], and it provides weak but non-trivial

evidence that the SAE features used in Chapter 3 are not arbitrary constructs but correspond to something the model actually uses.

Two limitations of this check. First, KL divergence is blunt: it captures that the output distribution shifted but not how, and cannot distinguish shifts aligned with the feature’s semantic interpretation from unrelated disruption. Behaviourally grounded causal validation would require a task-specific readout (e.g. change in refusal rate on JailbreakBench under safety-feature steering, or instruction-following accuracy under instruction-feature steering), which was not implemented. Second, the check tests only the existence of a structural difference between real and random directions, not its semantic meaning. The result is reported as a weak structural observation and not interpreted as evidence for or against the wrapper hypothesis. The full activation-steering experiment that would provide a causal complement to Chapter 3 is discussed in Subsection 5.5 as the most important future work.

4.4 Summary of Checks Performed

Chapters 2 and 3 report four independent robustness mechanisms, each targeting a different possible failure mode. They are collected here for convenience.

First, the classification is subjected to a 25-point threshold sensitivity grid (§3.1), confirming that the qualitative shape (preserved minority, modified dominant, exclusive classes sizeable but not behaviourally central) is robust across the grid. The primary operating point (0.7, 0.3) is representative, not cherry-picked.

Second, the labelling vocabulary is subjected to inter-annotator reliability via dual-LLM labelling with identical prompts and schemas. The 60% agreement rate and structural concentration of disagreement at specific boundaries constitute a rater-disagreement diagnostic reported as a finding. Every distributional claim in Sections 3.2–3.3 uses category counts rather than individual labels in consequence.

Third, the targeted analysis uses three probe datasets to run two behavioural contrasts—safety (JailbreakBench) and instruction-following (Alpaca [33] versus TruthfulQA as factual baseline)—rather than a single contrast, with Welch’s t -tests at $p < 0.01$ and a minimum activation filter. The convergence of those two contrasts on the same qualitative reading is the

cross-contrast robustness check.

Fourth, the cross-SAE direction check (§4.3) finds a consistent structural gap between real and random directions, supporting SAE features as non-arbitrary axes of variation.

The descriptive conclusions of Chapter 3 are defended by four robustness checks: threshold stability, inter-annotator reliability, cross-contrast convergence, and real-versus-random structural distinctiveness. Each targets a different objection and each is reported with enough detail to reproduce. None substitutes for a behaviourally grounded causal experiment, and the absence of such an experiment is the most important limitation of this thesis, named as the primary future work in Subsection 5.5.

5 Conclusion

This chapter returns to the motivation of Subsection 1.1 and the research question of Subsubsection 1.2.4, states the conclusion the evidence supports, and develops its implications for alignment practice, interpretability methodology, and future work. Section 5.1 states the conclusion at the level of confidence the data warrants. Section 5.2 maps the findings onto the questions that originally framed the project. Section 5.3 draws out implications for alignment. Section 5.4 reframes the pipeline as a reusable methodological contribution. Section 5.5 identifies future work, with behaviourally grounded causal validation as the top priority. Section 5.6 closes with a brief reflection.

5.1 The Conclusion, Stated at the Level the Data Supports

The Chapter 3 evidence supports the following conclusion in descriptive and correlational terms. At layer 13 of Gemma 3 1B, the features most behaviourally relevant to instruction-following and safety-related prompts are predominantly reorganised pre-existing features rather than newly introduced ones. The strongest version of this (that instruction tuning introduces no new features at all) is not supported: the IT-exclusive class is not empty, and a non-trivial fraction of active IT features lack a strong counterpart in the PT dictionary. What the evidence does support is the weaker claim that the features carrying the behavioural signatures of instruction tuning (responding to instructions and to harmful content) are not in the IT-exclusive class but in the modified class, whose members by construction have counterparts in the base model. The IT-exclusive class exists; it is not where the behavioural action is.

This conclusion is consistent with the wrapper hypothesis of Jain et al. [13], originally formulated in synthetic settings using pruning and linear probing. This thesis extends that hypothesis, for the first time to the author’s knowledge, to a real instruction-tuned language model at the feature level, using SAEs as the descriptive vocabulary and targeted differential analysis using three probe datasets to instantiate two behavioural contrasts as the behavioural grounding. The extension is not causal confirmation (Chapter 4 is explicit about this) but it is a substantive descriptive result linking the Jain et al. proposal to a production-class model. Two

findings bear the weight of this conclusion: the cross-model feature-index overlap in the §3.3 instruction-following contrast, where top-ranked differential features in the IT model coincide with top-ranked features in the PT model with order-of-magnitude effect sizes in both; and the distributional pattern across Sections 3.1–3.2, where instruction-style features concentrate in the modified class rather than either the preserved or the IT-exclusive class.

The conclusion is also consistent with the broader mechanistic fine-tuning work since Jain et al.: Prakash et al. [28] report that fine-tuning enhances existing mechanisms for entity tracking rather than installing new ones, and the sparse-model-diffing literature has converged on similar findings in other model families. This thesis adds a data point at the feature level, in a real instruction-tuned LLM, using matched-SAE comparison. Agreement across independent settings and tools is weak but non-trivial evidence that the wrapper reading is not an artefact of any single methodology.

5.2 Return to the Motivation and the Research Question

Subsection 1.1 motivated this thesis on the grounds that post-training is where most user-facing behaviour comes from, and that the mechanism of instruction tuning is contested between a “superficial alignment” reading (fine-tuning adjusts style and format over base-model capabilities [38]) and a “new capability” reading (fine-tuning installs genuinely new computational structure). Subsubsection 1.2.4 narrowed this to a specific research question: does instruction tuning introduce new features or reorganise existing ones? The answer, stated in §5.1, falls on the reorganisation side without going all the way to “no new features at all.”

This connects back to the superficial-alignment literature in a specific way. Zhou et al.’s LIMA observation was that a small amount of high-quality supervised data can produce strong instruction-following behaviour, implying that the substrate must already exist in the base model. The feature-level evidence here is consistent with that interpretation in a way earlier work could not be: the substrate is directly visible as specific SAE feature indices already selective for instruction-shaped prompts in the base model, whose selectivity is amplified by instruction tuning. Features 2006, 446, and the others in the cross-model overlap set of §3.3 are concrete representational objects that the superficial-alignment literature argues must exist,

shown here to actually exist and to be individually identifiable.

The safety contrast tells a weaker but parallel version of the same story, and it matters for the alignment community’s concern with how safety behaviours arise. The §3.3 finding that PT and IT models share top differential features on harmful prompts, and that no feature in the IT top-20 safety list belongs to the IT-exclusive class, implies that the representational basis for safety-relevant signals is already present in the base model as selectivity of knowledge-semantics and instruction-style features for harmful content. Instruction tuning reweights which base-model features are most active on harmful prompts but does not create a dedicated safety cluster from nothing. This is a more specific version of Jain et al.’s claim that fine-tuning installs a “minimal transformation ... on top of the underlying model capabilities” [13]: here the minimal transformation is a reweighting of activation patterns over a pre-existing feature basis rather than a change to the basis itself.

5.3 Implications for Alignment Practice

If the descriptive picture of §5.1 holds more broadly than this case study (a hypothesis this thesis motivates but cannot prove), several implications for alignment practice follow. They are stated here as hypotheses, not as claims established by the current evidence.

Base-model auditing becomes a meaningful safety intervention. If instruction tuning primarily reorganises existing features, the base model’s feature inventory sets an upper bound on what the instruction-tuned model can represent. A feature absent from the base model cannot be reweighted into existence; conversely, a feature present in the base model, whether dormant or latent, becomes a candidate for amplification under fine-tuning. An alignment team concerned about what behaviours an instruction-tuned model might exhibit therefore has a principled reason to audit the base model’s SAE features for latent capabilities fine-tuning might surface. This is a specific version of the jailbreak concern [40] that behaviours absent from the IT model’s typical outputs may still be recoverable through adversarial prompting, and of the sleeper-agent concern [12] that safety training may mask rather than remove harmful capabilities. The feature-level evidence here supports both in a narrower form: the capabilities fine-tuning operates on are already in the base model, so adversarial prompting that bypasses

fine-tuning should surface base-model features, not instruction-tuning-specific ones.

Safety fine-tuning is a reweighting, not a replacement. The §3.3 finding that IT safety features are substantially drawn from the base-model population implies that safety behaviours are carried by modulations of pre-existing features, not a dedicated circuit added during fine-tuning. This is consistent with the observation [14] that safety fine-tuning can be undone by further fine-tuning on superficially unrelated tasks: if safety is a reweighting rather than a structural addition, any subsequent training that perturbs those weights (in directions that need not be adversarial) can plausibly disrupt it. The feature-level picture does not prove this mechanism, but it is the kind of evidence such a mechanism would imply.

Cross-checkpoint feature transfer is more principled than arbitrary-direction transfer. The §4.3 robustness check found that SAE decoder directions learned on one checkpoint produce structurally distinguishable effects from random directions of matched magnitude when injected into the other checkpoint. This is weak in that it does not validate any specific behavioural steering claim, but non-trivial in suggesting the Gemma Scope 2 [19] feature basis is not arbitrary from the model’s perspective: the model processes these directions differently from random perturbations. For practitioners considering using SAE features trained on one checkpoint to steer, probe, or monitor another, this is weak permission to do so, with the caveat that behavioural steering claims require task-specific validation.

5.4 The Pipeline as a Reusable Methodological Contribution

Beyond its empirical findings, the thesis contributes a pipeline applicable to any model pair with matched SAEs. The four stages (activation collection, four-way classification with threshold sensitivity, dual-LLM consensus labelling, targeted differential analysis) are designed for independent reuse. A researcher interested only in classification needs the activation collection and Phase 2 matching code; the behavioural analysis of Phase 4 can be applied to features identified by any means; the dual-LLM labelling is model-agnostic.

Two methodological choices are contributions in their own right. First, the four-way classification using two complementary similarity measures (decoder cosine and activation correlation) is more informative than single-measure classifications. The observation that the modified

class dominates across the entire threshold grid is the classification’s main yield: the dominant mode of change between checkpoints is gradual reshaping of an existing feature basis, not replacement or preservation. A single-threshold single-measure classification cannot produce this kind of distributional framing. Second, the dual-LLM labelling treats annotator disagreement as a first-class observable rather than noise. The finding that disagreement concentrates at the `language_syntax` $\leftrightarrow$ `instruction_style` boundary turns an apparent weakness of automated labelling into a measurement of how fuzzy the taxonomic boundaries actually are. Single-annotator SAE labelling work would miss this entirely.

Together these choices define a template for matched-SAE comparison studies. Gemma Scope 2 [19] made the Gemma 3 1B comparison possible; a future release providing matched SAEs for a larger model, an RLHF-trained variant, or a non-English model would immediately enable the same pipeline to produce directly comparable results. The main methodological contribution is not the specific Gemma 3 1B finding but the template that made it possible and that can be reused as matched SAEs become more widely available.

5.5 Future Work

Six directions of future work follow from the findings and limitations, in roughly decreasing order of importance.

Behaviourally grounded causal validation. The most important limitation (Chapter 4) is the absence of a causal experiment testing whether candidate features actually drive the behaviours they are selective for. A full version would steer individual features (e.g. cross-model instruction-style features 2006 and 446, or safety-relevant features 262 and 2184) and measure task-specific readouts: refusal rate on JailbreakBench under safety-feature steering, or instruction-following accuracy on Alpaca under instruction-feature steering. The KL-based readout of §4.3 is inadequate because it measures distributional shift without measuring whether the shift is task-meaningful. This experiment should be prioritised above all other future work.

Multi-layer replication. Chapter 3 reports results only at layer 13. Re-running the pipeline at every layer with matched SAEs would produce a layer-by-layer picture of where instruction tuning exerts its effects. The layer profile could distinguish between instruction tuning acting

uniformly across the residual stream, concentrating at middle layers (where semantic features are richest), or concentrating at output layers (where behavioural decisions are made).

Cross-family replication. These findings are a single case study. Replicating on other model families as matched SAEs become available would test whether the wrapper-consistent picture holds beyond Google DeepMind’s instruction tuning of Gemma 3 1B. Llama 3, Mistral, and Qwen are plausible targets; the bottleneck is matched-SAE availability, not pipeline complexity.

Near-boundary safety prompts. Rerunning the targeted differential analysis on prompts near the refusal boundary—where small wording changes flip compliance—would test whether the apparent absence of dedicated safety features in the IT top lists survives a stronger behavioural probe than the fixed harmful set used in §3.3.

Cross-checkpoint crosscoders. Replacing the two separate SAEs with a single crosscoder trained jointly on both PT and IT residual streams, as in the sparse-model-diffing literature, would sidestep the matched-SAE comparison problem by design. Crosscoders were not used here because the Gemma Scope 2 [19] crosscoder releases are cross-layer rather than cross-checkpoint, and no PT↔IT crosscoder was publicly available. A future release would allow the classification to be re-derived from a single shared dictionary, and comparing separate-SAE and crosscoder results would be methodologically interesting.

Active-feature threshold sensitivity. A minor unfinished piece: no formal sensitivity grid was computed over the 0.1% active-feature cutoff. Informal checks suggested classification proportions are not strongly sensitive within the 0.05%–0.5% range, but extending the §2.4 grid to include this threshold as a third axis would close the loose end.

5.6 Closing Reflection

This thesis set out to test a contested hypothesis about instruction tuning using a feature-level pipeline, and it reports a descriptive answer consistent with the wrapper reading: at layer 13 of Gemma 3 1B, the features most behaviourally relevant to instruction-following and safety prompts are predominantly reorganised pre-existing features rather than newly introduced ones. The answer is not causal, not multi-layer, not multi-model, and not the last word. It is a con-

crete, reproducible data point on the feature-level side of a debate that has so far been conducted mostly at the behavioural and circuit level, and a pipeline that can be re-run as matched SAEs become available for other model pairs. The wrapper hypothesis was originally formulated for synthetic tasks and tested with pruning and probing; this thesis extends it to a production-class instruction-tuned model and tests it with dictionary learning. The convergence of independent methods on compatible readings is, in the author's view, the most significant contribution of this thesis, and the next steps (causal validation, multi-layer replication, cross-family replication) are the natural continuation of the line of work it joins.

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